MOBILE COMPUTING

ASSIGNMENT 1 – CO1

DESIGN, ANALYSE AND OPTIMISE A HEXAGONAL CELLULAR NETWORK

FOR SPECTRUM ALLOCATION, INTERFERENCE MANAGEMENT AND SCALABLE COVERAGE PLANNING IN A UNIVERSITY TOWNSHIP

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ASSIGNMENT OVERVIEW

This assignment develops a frequency-planning and capacity design for a cellular network serving a 12 km² university township. The design uses a fixed allocation of 336 duplex voice channels, a busy-hour traffic intensity of 0.025 Erlang per subscriber and a target call-blocking probability not greater than 2%.

The report compares fixed frequency reuse, sectorised reuse and dynamic channel assignment. It then examines capacity, interference, scaling, repeated cell splitting, simulation methodology and a temporary response to a large stadium event.

Design assumptions

Parameter	Adopted value / assumption
Township area	12 km²
Available duplex traffic channels	336 (control-channel overhead already excluded)
Subscriber traffic	0.025 Erlang per subscriber in the normal busy hour
Target grade of service	2% call blocking
Minimum SIR design target	18 dB for the conservative planning case
Path-loss exponent	n = 3.0 for general planning; sensitivity checked at 2.5 and 4.0
Initial reuse cluster	N = 7
Illustrative cell radius	R = 0.5 km
Cell geometry	Regular hexagonal cells; idealised equal-radius model
Important limitation	Exact site count and Erlang-B capacity depend on terrain, channel grouping and measured traffic; values here are planning estimates.

Traffic forecast

Subscribers	Offered traffic (Erlangs)	Interpretation
1,000	25	Initial deployment
10,000	250	Growing campus usage
20,000	500	Expanded township
100,000	2,500	Long-term high-demand case
25,000 event attendees	1,250 additional Erlangs	Temporary stadium surge: 25,000 × 0.05

Offered traffic is calculated as $A = S \times a$, where S is the number of subscribers and a is the average traffic per subscriber. These values are offered traffic, not automatically carried traffic; carried traffic is limited by the available channels and the selected blocking target.

TASK 1 – DESIGN ALTERNATIVE FREQUENCY-PLANNING STRATEGIES

Three different strategies are considered. They share the same fixed spectrum but use it in different ways to manage capacity and interference.

Strategy 1: Fixed cluster frequency reuse

The service area is divided into hexagonal cells. The 336 channels are divided into N channel groups, and each group is assigned to one cell in a repeating cluster. Cells using the same group are separated by a reuse distance. A larger cluster generally improves co-channel interference but reduces the number of channels available per cell.

- Access concept: the available duplex channels may be implemented using FDMA or time slots in TDMA. Control channels carry setup and signalling; traffic channels carry voice calls.
- Design principle: spatial reuse. The same frequency group is reused in sufficiently separated cells.
- Preprocessing: township traffic map, propagation survey, hexagonal grid, candidate site locations and channel plan. The operator bears the survey, planning and deployment cost.

Iterative procedure:

1. Choose candidate integers i and j and calculate $N = i^2 + ij + j^2$.
2. Reject cluster sizes that do not meet the selected interference target.
3. Divide 336 channels into N groups as evenly as possible.
4. Assign each group to a cell in the repeating hexagonal pattern.
5. Estimate traffic and blocking in each cell; adjust cell radius or cluster choice if required.

Pseudocode: FOR each valid (i,j): calculate N; calculate $Q = \sqrt{(3N)}$; estimate SIR; IF SIR meets target, allocate 336/N channels per cell and evaluate blocking; retain feasible designs; END.

Strategy 2: Sectorised frequency reuse

Each base station is divided into directional sectors using antennas, commonly three 120° sectors or six 60° sectors. Directional antennas reduce the number of strong co-channel interferers seen by a sector. The same channel allocation can therefore achieve better SIR than an omni-directional cell, subject to antenna alignment and side-lobe effects.

- Access concept: FDMA/TDMA channel groups are assigned to sectors; control and traffic channels are maintained for each sector according to the radio system.
- Design principle: spatial filtering. Antenna directionality suppresses unwanted signals from other directions.
- Preprocessing: antenna orientation and height survey, sector coverage prediction, traffic distribution by direction and field verification. The operator bears the engineering and installation cost.

Pseudocode: create sectors for every site; assign channel groups; calculate desired signal and sector-weighted interference; check SIR and blocking; rotate or adjust antenna parameters; keep the lowest-cost feasible layout.

Strategy 3: Dynamic channel assignment and channel borrowing

Instead of permanently fixing every channel to one cell, a cell requests a free channel when a call arrives. The controller checks whether the channel is available and whether its use would create unacceptable interference in neighbouring cells. A channel may be borrowed from a lightly loaded neighbour when the interference constraints remain satisfied.

- Access concept: channels are selected from a shared pool. Control signalling coordinates channel requests, reservations and releases; traffic channels carry active calls.
- Design principle: trunking efficiency and interference-aware resource allocation.
- Preprocessing: live or regularly updated traffic measurements, neighbour-cell interference information, channel occupancy records and a controller. The operator bears the measurement, software and signalling cost.

Pseudocode: when a call arrives, list free channels; rank candidates by interference and reuse constraints; assign the first feasible channel; if none is available, request borrowing from a neighbour; if no safe channel exists, block the call; release the channel when the call ends.

Comparison

Strategy	Main strength	Main limitation	Governing principle
Fixed reuse	Simple, predictable and easy to plan	Rigid allocation can leave channels idle in low-traffic cells	Spatial reuse
Sectorised reuse	Improves SIR by reducing interferers	Antenna cost, sector-edge gaps and more complex handoffs	Spatial filtering
Dynamic assignment / borrowing	Adapts to uneven and changing traffic	Needs coordination and reliable interference measurements	Trunking and interference control

TASK 2 – IDENTIFY THE GOVERNING DESIGN PARAMETER

Strategy	Dominant parameter	Why it governs performance
Fixed reuse	Cluster size N and reuse ratio Q = D/R	N determines channels per cell and the separation between co-channel cells.
Sectorised reuse	Number of sectors and effective interferer count i_0	More directional sectors can reduce the number and strength of dominant interferers.
Dynamic assignment	Available-channel pool and channel	The controller's choice depends on which channels are free and safe at

	occupancy / blocking	the time of a request.

As subscribers increase from 1,000 to 100,000+, fixed reuse cannot create additional spectrum; it must improve spatial reuse through smaller cells and careful frequency planning. Sectorisation can improve SIR and reuse efficiency, while dynamic assignment can redistribute unused capacity between cells. These methods improve utilisation but do not remove the finite-spectrum limit.

TASK 3 – DERIVE THE CAPACITY AND INTERFERENCE FUNCTIONS MATHEMATICALLY

Hexagonal geometry and reuse

For a regular hexagonal cell with cell radius R (centre to vertex), the area is $A_{cell} = (3\sqrt{3}/2)R^2$.

For a cluster formed by integers i and j, the cluster size is $N = i^2 + ij + j^2$. The co-channel reuse distance is $D = R\sqrt{(3N)}$, and the reuse ratio is $Q = D/R = \sqrt{(3N)}$.

For the adopted N = 7 and R = 0.5 km: $Q = \sqrt{21} = 4.583$; $D = 0.5 \times 4.583 = 2.291$ km.

Cell area = $(3\sqrt{3}/2)(0.5^2) = 0.6495$ km². Estimated cells to cover 12 km² = 12/0.6495 = 18.48, therefore approximately 19 cells before adding overlap, boundary and capacity margins.

Channels per cell

With 336 duplex traffic channels and N = 7, the average allocation is 336/7 = 48 channels per cell in each reuse group. If the division is not exact, groups must be allocated as integer channel counts whose total remains 336.

A cluster contains seven cells, so each frequency group is reused once per cluster. The 19-cell township estimate therefore contains about 19 cell areas, but not 19 unique channel groups; reuse is what allows the same 336-channel allocation to serve a larger area.

Erlang-B traffic capacity

For C traffic channels and offered traffic A, the Erlang-B blocking probability is:

B(C,A) = [A^C/C!] / Σ(k=0 to C)[A^k/k!].

The carried traffic is $A_{carried} = A(1 - B)$. The design requirement is $B(C,A) \leq 0.02$. The exact supported traffic for 48 channels should be obtained by solving the Erlang-B equation numerically or using an Erlang table; the following Python appendix includes a stable recursive calculation.

For a cell with C = 48, the maximum offered traffic is the value $A_2\%$ satisfying $B(48,A_2\%) = 0.02$. The township capacity estimate is approximately the sum of the supported traffic across its cells, while actual capacity also depends on uneven traffic, handoff reserves and interference restrictions.

Co-channel SIR

A commonly used first-tier approximation is SIR ≈ (D/R)^n / i_0 = (√(3N))^n / i_0, where n is the path-loss exponent and i_0 is the number of dominant first-tier co-channel interferers. In an omni-directional hexagonal layout, i_0 is often approximated as six.

For N = 7, n = 4 and i_0 = 6: SIR = $21^2/6$ = 73.5; in decibels, $10\log_{10}(73.5)$ ≈ 18.66 dB. This is consistent with the assignment's stated sanity-check value of about 18.7 dB.

For n = 3, the same approximation gives SIR = 21^1.5/6 ≈ 16.04, or about 12.05 dB. This demonstrates why the path-loss environment matters. Sectorisation changes the effective number and strength of interferers; a simple model can represent this by replacing i_0 with an appropriately reduced effective interferer count, but real antenna patterns should be measured or simulated.

Planning cost and cost–capacity relation

A simple planning-cost model is C_total = C_survey + N_sites × C_site + C_controller + C_antenna + C_optimisation. A useful comparison metric is cost per supported Erlang: C_total / A_carried, provided the same cost boundary is used for every strategy.

Smaller R increases the number of sites approximately in proportion to $1/R^2$. This can increase capacity through spatial reuse, but it also raises site, backhaul, power, maintenance and handoff costs. Therefore, the best design is not necessarily the design with the most cells.

TASK 4 – ANALYSE BEST, AVERAGE AND WORST-CASE BEHAVIOUR

Mobile position and SIR

Case	Mobile position	Expected radio condition	Reason
Best	Near serving base station	High desired signal and usually high SIR	The desired path is short; interfering sites are relatively distant.
Average	Typical mid-cell location	Moderate SIR	Desired and interfering path losses are more balanced.
Worst	Cell edge or corner, especially near a co-channel boundary	Low SIR and possible outage	The desired signal is weak and one or more co-channel interferers may be relatively close.

The exact SIR at a location is SIR = $P_0 d_0^{-n} / \Sigma(P_i d_i^{-n})$, where d_0 is distance to the serving site and d_i are distances to co-channel interferers. This equation should be evaluated for actual site coordinates to obtain numerical best, average and worst values. The first-tier formula in Task 3 is an idealised reference, not a substitute for a site-specific map.

Traffic and blocking

- Off-peak: low offered traffic, few occupied channels and normally very low blocking.
- Average period: moderate channel occupancy and moderate blocking risk.
- Busy hour: highest normal offered traffic; the 2% blocking target is checked here.
- Event surge: the stadium's concentrated demand can exceed local channel capacity even when township-wide average capacity appears adequate.

Shadowing can temporarily weaken the serving signal or strengthen an interferer. A higher path-loss exponent reduces distant interference more rapidly, which can improve reuse SIR in some layouts; however, it can also increase coverage holes and make cell-edge links more sensitive. A lower exponent allows interference to travel farther and can make frequency reuse more difficult.

For numerical analysis, evaluate the SIR equation at a dense set of mobile locations and calculate the 5th percentile, median and 95th percentile. For blocking, calculate Erlang-B separately for each cell and time period using the traffic assigned to that cell.

TASK 5 – SCALING AND SENSITIVITY ANALYSIS

Quantity	Approximate relationship	Meaning
Channels per cell	$C_{cell} \approx 336/N$	Increasing N reduces channels assigned to each cell.
Reuse distance	$D = R\sqrt{(3N)}$	Larger N increases co-channel separation.
SIR	$SIR \propto N^{(n/2)} / i_0$	Larger N generally improves SIR in the idealised model.
Number of cells	$N_{cells} \propto 1/R^2$	Halving radius requires about four times as many cells for the same area.
Offered traffic	A = subscribers × 0.025	Traffic grows linearly with subscriber count in the normal busy hour.

At 1,000 subscribers the offered load is 25 Erlangs. At 10,000 it is 250 Erlangs; at 20,000 it is 500 Erlangs; and at 100,000 it is 2,500 Erlangs. A fixed seven-cell reuse pattern does not automatically provide enough capacity for the largest forecast. Smaller cells, sectorisation and dynamic assignment should be introduced where measured demand requires them.

When n increases from 2.5 to 4, distant co-channel signals decay more rapidly, which can improve reuse isolation. However, dense buildings also create shadowing, penetration loss and uneven coverage. A practical break-even point occurs when the busy-hour load of a cell approaches its Erlang-B limit or when the measured SIR/outage target fails. At that point, sectorisation, traffic offload or cell splitting should be considered.

TASK 6 – FORMULATE THE RECURRENCE FOR REPEATED CELL SPLITTING

In uniform splitting, each parent cell is divided into four smaller cells. The radius is halved at each round: $R_k = R_0/2^k$. The number of cells after k rounds is:

$M_k = 4M_{(k-1)}$, with M_0 equal to the initial number of cells; hence $M_k = M_0 \times 4^k$.

If each new cell can support approximately the same traffic as the original cell under the same channel plan, aggregate offered-capacity opportunity scales as $K_k = K_0 \times 4^k$. This is a geometric planning approximation; interference, spectrum reuse, backhaul and handoff limits must still be checked.

For selective splitting, let f_k be the fraction of cells split at round k. Each split replaces one cell with four, adding three cells. Therefore $M_k = M_{(k-1)}(1 + 3f_k)$. If f is constant, $M_k = M_0(1 + 3f)^k$.

The transmit-power relation requested in the assignment is $P' = P/2^n$ for the specified splitting procedure. More generally, with path-loss exponent n and radius halved, the power needed to maintain the same received power at the new cell edge scales as $P_k = P_0/2^{(kn)}$, subject to the link budget and system power-control rules.

Handoff load increases as cell radius decreases because users cross more cell boundaries per unit distance. A simplified relation is that boundary-crossing frequency is proportional to user speed divided by cell dimension, so it rises approximately as $1/R$.

TASK 7 – SOLVE THE RECURRENCE AND INTERPRET THE RESULT

Let D be the demand to be supported and C_0 the supported capacity of the initial design. Under the idealised uniform four-way split, capacity after k rounds is $C_k = C_0 \times 4^k$.

The requirement is $C_0 \times 4^k \geq D$. Dividing by C_0 gives $4^k \geq D/C_0$. Taking logarithms gives $k \geq \log_4(D/C_0)$. Since k must be a whole number:

$k_{min} = \max(0, \lceil \log_4(D/C_0) \rceil)$.

For selective splitting with constant split fraction f, capacity is approximated by $C_k = C_0(1 + 3f)^k$, so $k_{min} = \max(0, \lceil \log(D/C_0)/\log(1 + 3f) \rceil)$. This is valid only when f is positive and the capacity-per-cell approximation is reasonable.

As an illustration, if demand becomes four times the initial capacity, one uniform split round is required. If demand becomes sixteen times the initial capacity, two rounds are required. Each round multiplies site count by four, reduces radius by half and increases handoff activity. The design must therefore balance capacity gain against site cost and mobility signalling.

TASK 8 – IMPLEMENT AND TEST THE PLANNING STRATEGIES

The following Python code provides a reproducible baseline for the mathematical checks. It calculates Erlang-B blocking, finds the offered traffic at a 2% target, estimates hexagonal geometry and computes the first-tier SIR approximation. It is a compact analytical simulator; a

full Monte-Carlo implementation should additionally generate mobile positions, site coordinates, antenna patterns and shadowing samples.

```
import math


def erlang_b(channels, traffic):
    """Erlang-B blocking probability using a stable recursion."""
    b = 1.0
    for c in range(1, channels + 1):
        b = (traffic * b) / (c + traffic * b)
    return b


def traffic_at_blocking(channels, target=0.02):
    """Binary search for the maximum offered traffic at target blocking."""
    low, high = 0.0, float(channels) * 2
    for _ in range(100):
        mid = (low + high) / 2
        if erlang_b(channels, mid) <= target:
            low = mid
        else:
            high = mid
    return low


def hex_area(radius):
    return (3 * math.sqrt(3) / 2) * radius ** 2


def cluster_size(i, j):
    return i*i + i*j + j*j


def reuse_distance(radius, n):
    return radius * math.sqrt(3*n)


def sir_db(n, path_loss, interferers=6):
    sir_linear = (3*n) ** (path_loss/2) / interferers
    return 10 * math.log10(sir_linear)


def offered_traffic(subscribers, erlangs_per_sub=0.025):
    return subscribers * erlangs_per_sub


if __name__ == "__main__":
    N = cluster_size(2, 1)  # N = 7
    R = 0.5
    channels_per_group = 336 // N
    area = hex_area(R)
    cells = math.ceil(12 / area)
    print("Cluster size:", N)
    print("Channels per group:", channels_per_group)
    print("Cell area (km^2):", round(area, 4))
    print("Estimated cells:", cells)
    print("Reuse distance (km):", round(reuse_distance(R, N), 3))
    print("SIR for n=4 (dB):", round(sir_db(N, 4), 2))
    print("2% traffic limit for 48 channels:",
          round(traffic_at_blocking(48), 2), "Erlangs")
    for users in [1000, 10000, 20000, 100000]:
        print(users, "users:", offered_traffic(users), "Erlangs")
```

Test plan:

6. Run the code with the same inputs for every strategy and retain the output.
7. Test cluster sizes N = 3, 4, 7 and 12, and path-loss exponents n = 2.5, 3 and 4.

8. Compare omni-directional and sectorised cases by changing the effective interferer model only when a defensible antenna pattern is available.
9. Use fixed random seeds in any Monte-Carlo extension so the strategies receive comparable mobile positions and shadowing samples.
10. Check the known case N = 7, n = 4 and six equal-distance interferers; the expected SIR is approximately 18.7 dB.

Note: this document does not claim that a full Monte-Carlo field simulation has been executed. The included code is reproducible and provides the baseline checks; measured simulation tables should be populated from the actual run before submission.

TASK 9 – RECORD AND PRESENT EXPERIMENTAL PERFORMANCE

The following measurements should be recorded for every strategy, subscriber load, cluster size and propagation condition:

- SIR distribution: mean SIR, median SIR and 5th-percentile SIR.
- Outage probability: percentage of sampled mobile locations below the selected SIR threshold.
- Blocking probability: Erlang-B value for each cell and aggregate busy-hour traffic.
- Resource and deployment measures: channels per cell, number of sites, sector count and handoff rate.
- Execution measures: simulation run-time and number of mobile samples.

Recommended plots are an SIR cumulative distribution function (CDF), blocking probability versus offered load, supported capacity versus cluster size, number of sites versus cell radius and cost per carried Erlang. Tables should use identical subscriber loads and propagation assumptions for all strategies.

Scenario	Offered traffic	Required recorded outputs
1,000 subscribers	25 Erlangs	SIR, blocking, occupied channels and site count
10,000 subscribers	250 Erlangs	Same measures; identify overloaded cells
20,000 subscribers	500 Erlangs	Same measures; test sectorisation and borrowing
100,000 subscribers	2,500 Erlangs	Identify required densification and capacity gap
Stadium event	1,250 additional Erlangs in 1 km²	Stadium-zone blocking, SIR and temporary capacity requirement

TASK 10 – COMPARE THEORY WITH SIMULATION

The analytical model predicts that increasing N improves the idealised SIR because co-channel cells are farther apart, but it reduces the number of channels assigned to each cell. Sectorisation

is expected to improve SIR by reducing the effective interference seen by a sector. Dynamic assignment is expected to reduce blocking in uneven traffic when safe channels are available in neighbouring cells.

A simulation may differ from theory for several reasons:

- The six-interferer approximation assumes equal distances and equal interference powers.
- Real layouts have finite numbers of cells and irregular township boundaries.
- Shadowing, building penetration and antenna side lobes change received powers.
- Mobile locations are not necessarily uniformly distributed; hostels, lecture halls and the stadium may form hotspots.
- Erlang-B assumes a particular loss-system traffic model with independent call arrivals and no waiting queue.
- Integer rounding means channels cannot always be divided equally between all groups.

The comparison should report the absolute difference between analytical and simulated SIR and blocking. The strategy that scales best is the one that meets SIR and blocking constraints at the lowest practical total cost, not simply the one with the largest theoretical capacity. Actual final ranking must be based on the completed simulation output.

TASK 11 – RECOMMEND AND DEFEND THE FINAL DESIGN

Recommended baseline: use a seven-cell fixed reuse pattern as the initial macro-cell plan, with three-sector antennas at high-traffic sites and interference-aware dynamic channel assignment for local load balancing. This hybrid approach retains a predictable frequency plan while improving spatial filtering and the use of temporarily idle channels.

- Initial planning values: $N = 7$, $R = 0.5$ km, $Q = 4.583$ and $D \approx 2.291$ km.
- The idealised six-interferer SIR is approximately 18.7 dB for $n = 4$, but only about 12.1 dB for $n = 3$. Therefore, the adopted 18 dB design target must be validated by propagation surveys and sector-level simulation.
- The 12 km² coverage estimate is approximately 19 hexagonal cells at the selected radius, before adding overlap, boundary and capacity margins.
- The 336-channel allocation gives an average of 48 channels per reuse group for $N = 7$. Per-cell blocking must be checked with Erlang-B against the 2% target.

This is a baseline recommendation rather than a claim that the network can support 100,000 subscribers without further deployment. At high demand, the operator must add capacity through selective cell splitting, micro-cells, sectorisation, offload and careful dynamic channel assignment. Costs include surveys, sites, antennas, backhaul, power, controller software and ongoing optimisation.

The design remains practical only if measured SIR, blocking, coverage and cost targets continue to be met. A fixed macro-cell design alone is not sufficient evidence of capacity for 100,000+ subscribers.

TASK 12 – DAY-4 DESIGN CHALLENGE: RE-EVALUATE AND ADAPT THE DESIGN

A. Impact analysis

The event brings 25,000 attendees into a 1 km² stadium zone. Each attendee generates 0.05 Erlang, so the additional offered traffic is 25,000 × 0.05 = 1,250 Erlangs over the event period. The corresponding event traffic density is 1,250 Erlangs/km².

The normal planning assumption is 0.025 Erlang per subscriber. The event attendee traffic per person is twice this normal per-subscriber value, and it is concentrated in one small zone. The assumptions of uniform traffic and steady demand no longer hold. The event lasts four hours and occurs about twice a month, so a temporary capacity response may be more economical than permanently redesigning the whole township.

B. Evaluation of the original design

The original macro-cell design may provide coverage, but the stadium's concentrated 1,250-Erlang surge can overload the cells serving the venue. Blocking is likely to rise first in the stadium cells because their local offered traffic can exceed the 2% Erlang-B capacity of the available channels. Interference may also increase if neighbouring cells borrow channels without adequate coordination.

The exact blocking probability cannot be stated without the number of channels allocated to the stadium zone, its reuse pattern and the traffic split among sectors. The correct evaluation is to insert the measured or modelled event load into Erlang-B for each stadium sector and compare the result with 0.02.

C. Adaptation or redesign

Recommended decision: MODIFY the design temporarily on event days rather than permanently redesign the township. Deploy temporary small cells or cells-on-wheels around the stadium, use directional 60° or 120° sectoring where appropriate, and enable controlled dynamic channel borrowing from lightly loaded neighbouring cells.

- Temporary micro-cells shorten the radio path and localise capacity to the 1 km² hotspot.
- Sector antennas help direct energy into the venue and reduce interference from other directions.
- Dynamic borrowing uses spare channels in surrounding cells, but only after checking co-channel interference and preserving a reserve for normal users.
- Transmit power should be adjusted to the smaller coverage footprint; the exact power setting requires a link-budget calculation.
- Handoff boundaries and neighbour lists must be tested before the event because smaller cells increase handoff frequency.

The temporary plan should be activated before the event, monitored during the four-hour period and removed or returned to standby afterwards. This avoids paying for permanent stadium-level densification throughout the month.

D. Evidence-based defence

The temporary modification is preferred because the event demand is large but localised and short-lived. Permanent township-wide cell splitting would increase site cost, power, maintenance and handoff load even on ordinary days. Temporary cells, sectorisation and controlled borrowing focus the extra capacity where and when it is needed.

The final decision must be confirmed using the event traffic model, Erlang-B calculations for each temporary sector, SIR simulation and a cost comparison between temporary deployment and permanent expansion. The event occurs roughly twice a month, so the operator should compare the recurring temporary deployment cost with the annualised cost of permanent sites before procurement.

CONCLUSION

This assignment examined how a fixed allocation of 336 duplex traffic channels can be reused across a 12 km² university township. Fixed cluster reuse provides a clear baseline, sectorisation improves spatial filtering, and dynamic channel assignment helps respond to uneven traffic. Hexagonal geometry gives the relationships between cell area, reuse distance and cluster size, while Erlang-B provides the blocking model for a specified number of traffic channels.

The recommended approach is a seven-cell reuse baseline supported by sectorisation and interference-aware dynamic assignment, followed by selective densification as demand grows. The event challenge is best handled initially through temporary stadium capacity rather than permanent township-wide redesign. Final deployment values must be validated with field measurements and completed simulation results.

REFERENCES

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APPENDIX A – DETAILED DESIGN CALCULATIONS

A.1 Offered traffic at each growth stage

The busy-hour offered traffic is calculated by multiplying the number of subscribers by the average traffic generated by one subscriber. The assignment specifies 0.025 Erlang per subscriber for normal conditions.

Subscriber count	Calculation	Offered traffic
1,000	1,000 × 0.025	25 Erlangs
10,000	10,000 × 0.025	250 Erlangs
20,000	20,000 × 0.025	500 Erlangs
100,000	100,000 × 0.025	2,500 Erlangs

An Erlang represents continuous use of one traffic channel for one hour. For example, 25 Erlangs may represent 25 channels occupied continuously for one hour, or a corresponding combination of shorter calls. Offered traffic describes demand before blocked calls are removed.

A.2 Hexagonal coverage calculation

The area of one ideal regular hexagonal cell is $A_{cell} = (3\sqrt{3}/2)R^2$. Using $R = 0.5$ km gives $A_{cell} = 0.6495$ km². The ideal cell count is $12/0.6495 = 18.48$, which is rounded upward to 19 cells.

This is an initial geometric estimate. A real network may need additional cells because of irregular township boundaries, buildings, terrain, overlap requirements, indoor coverage and non-uniform demand. A radio survey is needed before fixing the final number of sites.

A.3 Reuse distance calculation

For a cluster size $N = 7$, the reuse ratio is $Q = \sqrt{(3N)} = \sqrt{21} = 4.583$. With a cell radius of 0.5 km, the reuse distance is $D = RQ = 0.5 \times 4.583 = 2.291$ km.

Reuse distance is the approximate separation between cells using the same frequency group. Increasing this distance can reduce co-channel interference, but a larger cluster also means fewer channels per cell because the fixed spectrum is divided among more groups.

A.4 First-tier SIR calculation

The simplified SIR equation is $SIR \approx (D/R)^n / i_0$. Since $D/R = \sqrt{(3N)}$, it becomes $SIR \approx (3N)^{(n/2)} / i_0$. For $N = 7$, $n = 4$ and $i_0 = 6$, the linear SIR is $21^2/6 = 73.5$.

Converting to decibels gives $SIR(dB) = 10\log_{10}(73.5) \approx 18.66$ dB. The result is an idealised estimate using six equal-distance interferers. Actual SIR depends on mobile position, antenna pattern, site geometry and propagation conditions.

APPENDIX B – FREQUENCY REUSE AND INTERFERENCE DISCUSSION

B.1 Why frequency reuse is required

Radio spectrum is limited, so a cellular operator cannot assign a completely different frequency to every cell in a large service area. Frequency reuse allows the same channel group to be used again in cells separated by a sufficient distance. This increases the number of users served by the same spectrum allocation.

The main challenge is co-channel interference. When two nearby cells use the same frequency, transmissions from one cell may interfere with reception in the other. The reuse plan must therefore balance capacity and signal quality.

B.2 Fixed reuse versus dynamic assignment

Fixed reuse is easy to plan because channel groups are assigned in advance. It offers predictable interference behaviour, but a cell may have unused channels while a neighbouring cell is overloaded. Dynamic assignment can use the shared channel pool more flexibly, although it requires control signalling and interference checks.

B.3 Role of sectorisation

Sectorisation divides a cell into directional coverage areas. A three-sector site commonly uses three 120-degree sectors, while a six-sector design uses six 60-degree sectors. Directional antennas focus energy toward intended users and can reduce the strength of signals arriving from unwanted directions.

Sectorisation does not create new spectrum. It improves the spatial use of the existing spectrum and may permit a more efficient reuse plan. The design must account for antenna side lobes, sector overlap, handoffs and installation cost.

B.4 Control channels and traffic channels

Control channels carry signalling information such as call setup, paging, channel assignment and handoff coordination. Traffic channels carry the actual user communication. The assignment states that the available 336 duplex voice channels are after control-channel overhead, so the capacity calculations in this report treat 336 as the usable traffic-channel allocation.

APPENDIX C – SIMULATION WORKFLOW AND DATA COLLECTION

C.1 Suggested workflow

- Create a hexagonal grid covering the 12 km² township and label each cell.
- Place candidate base stations at the cell centres and define neighbouring cells.
- Generate frequency groups for the selected cluster size.
- Generate mobile-user positions using the same random seed for each strategy.
- Calculate serving-signal power and co-channel interference for every mobile position.

- Calculate SIR and classify locations below the selected threshold as outage locations.
- Calculate offered traffic per cell and use Erlang-B to estimate blocking probability.
- Repeat the experiment for the subscriber forecasts and propagation exponents.
- Export the measurements to tables and plots for strategy comparison.

C.2 Suggested experimental record sheet

Run	Subscribers	Strategy	N / sectors	Path-loss n	Mean SIR	5th percentile SIR	Blocking	Outage	Sites
1	1,000	Fixed reuse	7 / omni	3.0	To be measured	To be measured	To be measured	To be measured	To be measured
2	10,000	Sectorised	7 / 3 sectors	3.0	To be measured	To be measured	To be measured	To be measured	To be measured
3	20,000	Dynamic assignment	As configured	3.0	To be measured	To be measured	To be measured	To be measured	To be measured
4	100,000	Hybrid	As configured	4.0	To be measured	To be measured	To be measured	To be measured	To be measured

The table is provided as a recording template. The entries marked 'To be measured' must be replaced with values produced by running the implementation. They should not be presented as measured results before the simulation is completed.

C.3 Validation checks

- Verify that the total number of allocated channels never exceeds 336.
- Verify that every active call receives one available traffic channel.
- Verify that the same random seed is used when comparing alternative strategies.
- Check that the N = 7, n = 4, six-interferer reference case produces approximately 18.7 dB.
- Check that blocking probability increases when offered traffic increases while the number of channels remains fixed.
- Check that a smaller cell radius increases the number of cells required to cover the same township area.

APPENDIX D – GLOSSARY OF IMPORTANT TERMS

Term	Meaning
Cell	A geographic area served by a base station or sector.
Cluster	A group of cells in which each frequency group is used once before the pattern repeats.
Frequency reuse	Using the same radio channels in geographically separated cells.
Co-channel interference	Interference caused by another transmitter using the

	same frequency channel.
Reuse distance (D)	Approximate distance between cell centres using the same frequency group.
Reuse ratio (Q)	Ratio of reuse distance to cell radius, Q = D/R.
Cluster size (N)	Number of cells in one frequency-reuse cluster.
SIR	Signal-to-interference ratio; desired signal power divided by interference power.
Erlang	A unit of traffic intensity representing continuous use of one channel.
Grade of service	A measure of service quality, often expressed as call-blocking probability.
Erlang-B	A formula used to estimate call blocking in a loss system with a fixed number of channels.
Sectorisation	Dividing a cell into directional antenna sectors.
Cell splitting	Replacing a large cell with smaller cells to increase spatial capacity.
Handoff	Transfer of an active connection from one cell or sector to another.
Path-loss exponent	A parameter describing how received signal power decreases with distance.

APPENDIX E – LIMITATIONS AND FUTURE IMPROVEMENTS

The calculations use a regular hexagonal layout and simplified propagation equations. These assumptions are useful for understanding the design principles but do not fully represent a real university township. Buildings, indoor users, terrain, road corridors and uneven population density can change both coverage and interference.

The first-tier SIR model assumes a fixed number of equal-distance interferers. A more accurate implementation should use actual base-station coordinates, antenna gain patterns, transmit powers, shadowing and mobile locations. The traffic model should also distinguish ordinary users, high-demand locations and event attendees.

Future improvements include a GIS-based coverage map, measured radio-propagation data, a more detailed Monte-Carlo simulator, automatic frequency assignment, real-time load balancing and a cost model that includes backhaul, power and maintenance. These improvements would make the proposed design more suitable for deployment planning.